



[Aptitude Multiple Choice Question – Rahul Kale (rahul@sunbeaminfo.com)]
[9881144174]



[Simple interest] Q1. Ram invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?

- A. Rs. 6400 B. Rs. 6500 C. Rs. 7200 D. Rs. 7500 E. None of these**

Difference between schemes A and B is $14\% - 11\% = 3\%$

$$13900 \times 11\% = 1529$$

$$2 \text{ YR} = 3508$$

$$1 \text{ YR} = 1754,$$

$$\text{extra interest} = 1754 - 1529 = 225$$

$$3\% = 225$$

$$\text{So, } 3 = 225 \times 100 \quad = 225/3 \times 100 == 7500$$

$$13900 - 7500 = 6400$$

Answer : A is correct 6400



[Profit / Loss] Q2. Nita purchased 20 dozens of toys at the rate of Rs. 375 per dozen. She sold each one of them at the rate of Rs. 33. What was her profit percentage?

- A. 3.5 B. 4.5 C. 5.6 D. 6.5

Rate per dozen = 375

total no. of dozens = 20

rate of one toy = $375 / 12$ (here 12 is for a dozen)

= Rs.31.25

Now, given selling price(S.P.) = 33

profit = (selling price – cost price) = $33 - 31.25 = 1.75$

Now , PROFIT % = $SP - CP / CP \times 100$

PROFIT % = $1.75 / 31.25 \times 100$

PROFIT % = $0.056 \times 100 = 5.6 \%$

Answer : C is correct 5.6 %

















[Percentage]

Q3. A fruit seller had some oranges. He sells 40% oranges and still has 420 oranges. Originally, he had:

- A. 588 oranges B. 600 oranges C. 672 oranges D. 700 oranges**

Let the number of oranges be x.

He sold = 40% of x

Left = 100-40 = 60% of x

$$60 / 100x = 420$$

$$60 / 20 / 100x / 20 = 420 \quad 3 / 5x = 420$$

$$X = (420 * 5) / 3$$

$$X = 2100 / 3$$

$$X = 700$$

Answer : D is correct 700

.



[Age]

Q4. What is John's present age, if after 10 years his age will be 5 times his age 5 years back.

- A. 6.25 years B. 7.75 years C. 8.75 years D. 10 years**

Let John's present age be x .

John's age before 5 years = $(x - 5)$

John's age after 10 years = $(x + 10)$

As per given condition, $(x + 10) = 5(x - 5)$

Solving the equation, we get $x + 10 = 5x - 25$

Solving the equation, we get $35 = 4x$

$$4x = 35$$

$$= 8.75 \text{ years}$$

Answer : C is correct 8.75 years

.



[Speed and Time] Q5. A boat covers 800 meters in 600 seconds against the stream and returns downstream in 5 minutes. What is the speed of the boat in still water?

- A. 1 m/s B. 1.5 m/s C. 2 m/s D. 2.5 m/s

$$\begin{aligned}\text{Speed upstream} &= \text{Distance} / \text{Time} \\ &= 800 / 600 = 8 / 6 \text{ m/s}\end{aligned}$$

$$\begin{aligned}\text{Speed downstream} &= \text{Distance} / \text{Time} \\ &= 800 / 5 \times 60 = 800 / 300 = 8 / 3 \text{ m/s}\end{aligned}$$

$$\text{Speed in still water} = \frac{1}{2} * (\text{speed downstream} + \text{speed upstream})$$

$$\text{Speed in still water} = \frac{1}{2} * (8 / 3 + 8 / 6)$$

$$\text{Speed in still water} = \frac{1}{2} * (8 + 16 / 6)$$

$$\text{Speed in still water} = \frac{1}{2} * (24 / 6)$$

$$\text{Speed in still water} = \frac{1}{2} * (4) = 2 \text{ m/s}$$

Answer : C is correct 2 m/s



[arrangements] Q6. Find the number of ways the letters of the word 'RUBBER' can be arranged?

A. 450

B. 362

C. 250

D. 180

The word 'RUBBER' contains 6 letters: 2 R , 2 B , 1 U , 1 E

Therefore, The required Number of ways:

$$= N! / (2R!) \times (2B!) \times (1U!) \times (1E!)$$

$$= 6! / (2 \times 1) \times (2 \times 1) \times (1) \times (1)$$

$$= 6 * 5 * 4 * 3 * 2 * 1 / (2) \times (2) \times (1) \times (1)$$

$$= 6 \times 5 \times 4 \times 3 \times 2 \times 1 / 4$$

$$= 6 \times 5 \times 3 \times 2$$

$$= 180$$

Answer : D is correct 180



[Calender] Q7. If the seventh day of a month is three days earlier than Friday, What day will it be on the nineteenth day of the month?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday

Given that seventh day of a month is three days earlier than Friday

So Seventh day is Tuesday

14th day is Tuesday

21st day is Tuesday

19th day is Sunday

Answer : A is correct Sunday



Q8. The difference between two numbers is 18. If four times the second number is less than three times the first number by 18, then what is the sum of these two numbers?

A. 100

B. 80

C. 86

D. 90

Let first number = x and second number = y

According to the question, $x - y = 18$...(i)

and $3x - 4y = 18$...(ii)

On multiplying Eq. (i) by 3 and then subtracting Eq. (ii) from it, we get

$$3x - 3y = 54$$

$$3x - 4y = 18 \quad (3x - 3y) - y = 18 \quad 54 - y = 18 \quad y = 54 - 18$$

$$y = 36$$

On putting the value of y in Eq. (i), we get

$$x = 18 + y = 18 + 36 \Rightarrow x = 54$$

Therefore, required sum = $x + y = 54 + 36 = 90$

Answer : D is correct 90

.



[Trains] Q9. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is:

- A. 1 : 3 B. 3 : 2 C. 3 : 4 D. None of these**

Let the speeds of the two trains be x m/sec and y m/sec respectively.

Then, length of the first train = $27x$ meters,

and length of the second train = $17y$ meters.

$$27x + 17y / x + y = 23$$

$$27x + 17y = 23 (x + y)$$

$$27x + 17y = 23x + 23y$$

$$27x - 23x = 23y - 17y$$

$$4x = 6y$$

$$x / y == 6 / 4$$

$$x / y == 3 / 2$$

Answer : B is correct 3 : 2



[Speed] Q10. A man takes 2.2 times as long to row a distance upstream as to row the same distance downstream. If he can row 55 km downstream in 2 hour 30 minutes, what is the speed of the boat in the still water?

- A. 40 km/hr B. 8 km/hr C. 16 km/hr D. 24 km/hr

Speed of the boat in downstream

= distance / time

$$= 55 / 2.5$$

$$= 55 \times 10 / 2.5 \times 10 = 550 / 25 = 22 \text{ km / hr}$$

Then, speed of the boat in upstream

$$= 22 / 2.2$$

$$= 22 \times 10 / 2.2 \times 10 = 220 / 22 = 10 \text{ km / hr}$$

∴ Speed of the boat in still water

$$= (22 + 10) / 2 = 32 / 2 = 16 \text{ km / hr}$$

Answer : C is correct 16 km / hr



Q11. In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green?

A. $1/3$

B. $3/4$

C. $7/19$

D. $8/21$

E. $9/21$

Number of red ball in a box = 8

Number of blue ball in a box = 7

Number of green ball in a box = 6

So the total balls in a box = $(8 + 7 + 6) = 21$ So the total number of outcomes = 21.

Now one ball is picked up randomly and we have to find the probability that the ball is neither red nor green. Now if the ball is neither red nor green than the remaining color left in the box is blue.

So the favorable number of outcomes are 7.

Now we know that the probability (P) is the ratio of favorable number of outcomes to the total number of outcomes.

So the probability (P) that the ball is neither red nor green is

$P = \text{Favorable number of outcomes} / \text{Total number of outcomes}$

$= 7 / 21$

$= 1 / 3$

Answer : A is correct $1 / 3$

.



[Cones]

Q12. If the ratio of volumes of two cones is 2 : 3 and the ratio of the radius of their bases is 1 : 2, then the ratio of their height will be :

- A. 3 : 4 B. 4 : 3 C. 3 : 8 D. 8 : 3**

let their radius be x and 2x, and their heights be h and H respectively Then,

$$(1 / 3 * \pi * x * x * \pi * h) / (1 / 3 * \pi * 2x * 2x * \pi * H) = 2 / 3$$

Remove common from both

$$= h / 4 H = 2 / 3$$

$$= h / H = 4 * 2 / 3$$

$$= h / H = 4 * 2 / 3$$

$$= h / H = 8 / 3$$

Answer : D is correct 8 : 3

.



[Average] Q13. The average of 99 items is 27. If 3 is added to each number of the series, the new average is

A. 24

B. 27

C. 33

D. 30

Average of 99 numbers is 27

3 added to each numbers

Average = sum / numbers

Sum= average * numbers

Sum= 99 * 27 = 2673

New total= 2673 + 99*3

New total= 2673 + 297

New total= 2970

new average= 2970 / 99

new average= 30

Answer : D is correct 30



[Average]Q13. The average of 99 items is 27. If 3 is added to each number of the series, the new average is

A. 24

B. 27

C. 33

D. 30

$$\text{New average} = (99 * 27) + (99*3) \quad / \quad 99$$

$$\text{New average} = 99 * (27 + 3) \quad / \quad 99$$

$$\text{New average} = 99 * (30) \quad / \quad 99$$

$$\text{New average} = (30)$$

Answer : D is correct 30



[Age] Q14. The present age of a father is three years more than three times the age of the son. Three years hence, father's age will be 10 years more than twice the age of the son. Determine their present ages.

A. 32 yrs

B. 33 yrs

C. 40 yrs

D. 35 yrs

Let the son's age be x Then , according to the condition, father's age $= 3x+3$

3 years later: Son's age $= x + 3$

Father's age $= 3x+3+3 = 3x+6$

According to the condition given ,

$$3x+6=10+2(x+3)$$

$$3x+6=10+2x+6$$

$$3x-2x = 10+6-6$$

$$\text{so } x = 10$$

Thus , the son's present age is 10 years and the father's age is

$$3x+3=3 \times 10+3= 30 +3 = 33 \text{ years.}$$

33 years

Answer : B is correct 33 years



Q15. A clock is begun at twelve. By 10 min. past 5, the hour hand has turned through:

A. 145°

B. 150°

C. 155°

D. 160°

In 12 hrs, hour hand takes one round i.e it moves by 360 degrees.

So in 1 hr, hour hand moves by 30 degrees.

In 10 min, hour hand moves by $30 \times 10 / 60$

In 10 min, hour hand moves by $300 / 60$

In 10 min, hour hand moves by 5 degrees

In 5 hr 10 min, hour hand moves by,

$30 \times 5 + 5$ degrees

$150 + 5$ degrees

155 degrees

Answer : C is correct 155 degrees



[Average]

Q16. The average marks of a class of 87 students is 56 When a new student was added and average becomes 56.5 Find marks of new student.

A 56

B 44

C 100

D 90

Average pf class is $87 * 56 = 4872$

New Average pf class is $88 * 56.5 = 4972$

Difference of averages = $4972 - 4872$

Difference of averages = 100

marks of new student is 100

Answer : C is correct 100

.



[profit or loss]

Q17. A book seller sells 84 books at the cost of 72 books find profit or loss?

A. 14.28%

B. 28.24%

C. 20.4%

D. 12.86%

To calculate the profit or loss percentage, compare the selling price with the cost price.

Cost price (CP) = Cost of 72 books

Let's assume the cost price of one book is "x. Then, the cost price of 72 books = 72x

The cost of 84 books = 84x The selling price (SP) for 84 books = 72x

Here $72x < 84x$ Hence, it will be a loss for the bookseller

$$\text{Loss} = \text{CP} - \text{SP} = 84x - 72x = 12x$$

To find the loss percentage, use the formula:

$$\text{Loss \%} = (\text{Loss} / \text{Cost Price}) \times 100$$

$$= (12x / (84x)) \times 100$$

$$= (1 / 7) \times 100$$

$$= 14.28 \%$$

The bookseller will get a 14.28% loss on selling 84 books at the cost of 72 books.

Answer : A is correct 14.28 %

.



Q18. The proportion of number of young men to that of the young ladies in a school is 3:2 .If 20% of young men and 25% of young ladies are grant holders, discover the percentage of the individuals who are not grant holders.

A. 64 %

B. 78 %

C. 84 %

D. 76 %

Let the no. of boys be 300 and the no. of girls be 200.

Boys holding scholarship = $20 / 100 * 300 = 60$

Girls holding scholarship = $25 / 100 * 200 = 50$

Total students holding scholarship = $60 + 50 = 110$

% of students not holding scholarship = $(500 - 110) / 500 \times 100$

% of students not holding scholarship = $(390) / 5 * 1$

% of students not holding scholarship = 78 %

Answer : B is correct 78%



[Time and Work]Q19. If 8 men working 9 hours a day can build a wall 18m long, 2m broad and 12m high in 10 days, how many men will be required to build a wall 32m long, 3m broad and 9m high by working 6 hours a day, in 8 days?

A. 30 B. 40 C. 50 D. 60

MEN	DAYS	Hour	L * B * H of wall
8	10	9	18 * 2 * 12 = 432
x	8	6	32 * 3 * 9 = 864

$$x = \frac{10 * 9 * 864 * 8}{8 * 6 * 432}$$

$$X = \frac{10 * 9 * 2 * 432 * 8}{8 * 6 * 432}$$

$$X = 180 / 6 = 30 \qquad \qquad \qquad 77760 / 2592 = 30$$

men will be required =30

Answer : A is correct 30

[Number System]Q20. A boy was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was:

A. 12

B. 13

C. 14

D. 15

Let the number be x.

$$52x - 25x = 324$$

$$27x = 324$$

$$x = 324 / 27$$

$$x = 12$$

The number to be multiplied was: 12

Answer : A is correct 12



[Alligation] Q10. The ratio, in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea when sold for Rs. 194.40 per kg, gives a profit of 20%
(Day1 10 th question 1 way)

A. 2 : 5 B. 3 : 5 C. 5 : 3 D. 5 : 2

Cost Price of first tea = Rs. 192 per kg.

Cost Price of Second tea = Rs. 150 per kg.

Mixture is to be sold in Rs. 194.40 per kg, which has included 20% profit. So,

Selling price of Mixture = Rs. 194.40 per kg.

Let the CP of Mixture be Rs. X per kg. Therefore, $X + 20\% \text{ of } X = \text{SP}$

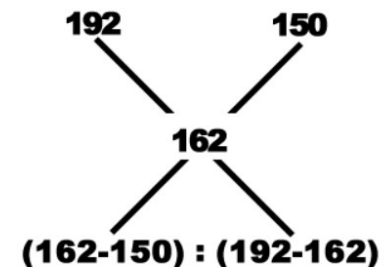
$$6x / 5 = 194.40 \quad 6x = 194.40 * 5 \quad 6x = 972 \quad x = 972/6 \quad x = 162$$

Let N kg of first tea and M kg of second tea to be added.

$$162 : 150 :: 192 - 162$$

$$N / M = 12 / 30 \quad (12/6) / (30/6) \rightarrow 2 : 5$$

Answer : A is correct 2: 5



[Alligation] Q10. The ratio, in which tea costing Rs. 192 per kg is to be mixed with tea costing Rs. 150 per kg so that the mixed tea when sold for Rs. 194.40 per kg, gives a profit of 20% (Day1 10 th question 2 way)

A. 2 : 5

B. 3 : 5

C. 5 : 3

D. 5 : 2

Cost Price of first tea = Rs. 192 per kg.

Cost Price of Second tea = Rs. 150 per kg.

Mixture is to be sold in Rs. 194.40 per kg, which has included 20% profit. So,

Selling price of Mixture = Rs. 194.40 per kg.

Selling price of Mixture = Cost Price * (100 + gain %) / 100

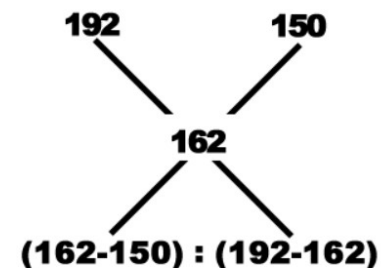
$$194.40 = \text{Cost Price} * (100 + 20) / 100$$

$$194.40 = \text{Cost Price} * (120) / 100$$

$$19440 = \text{Cost Price} * 120$$

$$\text{Cost Price} = 19440 / 120 = 162$$

Answer : A is correct 2: 5



$$12 : 30$$

$$2 : 5$$

